

StreamShark:
relying on
Google Cloud
to stream
videos
featuring
heads of state
and CEOs to
millions of
people

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StreamShark 

About StreamShark

Formerly MetaCDN, [StreamShark](https://streamshark.io/)
(<https://streamshark.io/>) provides live
streaming and video on demand services via an
end-to-end platform. The business focuses on
making tools and workflows for video stream
operators and marketing teams within client
businesses—enabling them to deliver content
and messaging in a seamless, high-quality and
high-performance way.

Industries: Technology

With Google Cloud
StreamShark delivers
reliable, high-quality
latency video streams
events involving
heads of state and

the ability to provide search, and other added services.

Location: Australia

Google Cloud results

- Delivers reliable, high-quality, low-latency video streams
- Meets cost requirements
- Gains ability to add interactive chat functionality

Scales to deliver video heads to millions of people worldwide

Datastore

Google Cloud (<https://cloud.google.com/>)

Firebase (<https://firebase.google.com/>), Cloud

App Engine (<https://cloud.google.com/appengine/>)

Google Workspace

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Datastore

(<https://cloud.google.com/datastore/>)

Firebase (<https://firebase.google.com/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

Google Workspace

(<https://workspace.google.com/>)

StreamShark (<https://streamshark.io/>)—formerly

MetaCDN—provides live streaming and video on demand services worldwide. “We founded MetaCDN in 2011 to deliver rich content and media to end users worldwide,” says James Broberg, founder & CEO at StreamShark. “We wanted to provide enterprise reliability and quality at prices smaller businesses can afford.” However, as the organization found video was the biggest driver of

growth, it began to tailor its products and solutions around the medium. MetaCDN started with on-demand before moving on to live streaming, and in September 2015 fully rebranded as StreamShark to capitalize on what Broberg describes as the insatiable demand for video.

“We operate an end-to-end platform for on-demand and live streaming of video content—in HD, 4K, 360VR, or other formats,” says Broberg. “Our focus is on making tools and workflows for video stream operators and marketing teams within client businesses. This enables them to get their content and messaging out to the world in a seamless, high-quality and high-performance fashion.”

Corporate communications at scale

While StreamShark delivers all kinds of video content, corporate communications at scale is an increasingly big slice of its business. “If you think of large organizations that have thousands to tens of thousands of employees around the world, they want to communicate culture, strategy, and business direction to people in offices all over the globe,” says Broberg. “They may do this via town halls or all-hands meetings featuring well-produced addresses by CEOs or other corporate leaders—and they will want employees to view that content in a high-quality, low-latency format, no matter where those employees are located.

“With StreamShark, audiovisual technicians can seamlessly pick up a video signal, send it to the

cloud, change it into required formats, and give it all the qualities they need to play back smoothly on all devices in all network conditions.”

International demand—particularly from United States corporations—is powering steep growth, with StreamShark’s business doubling in size year-on-year over most of the last four or five years.

In 2011, with a small launch budget and three-person team, the company elected to start operations in the cloud. “Our approach necessitated a cloud-first strategy,” says Broberg. “For example, we didn’t have the resources to assign someone to infrastructure and operations. Effectively outsourcing to the cloud would allow us to focus on business logic and applications.” Running in the cloud would also enable StreamShark to align spend with scale and synchronize with customer demand and growth in revenue.

“Accessing infrastructure close to our customer was very important to us—making selection of a global cloud provider a no-brainer,” adds Broberg. “Cost was important too—we wanted some economies of scale as we grew.

“Finally, we wanted our risk—and cost—to remain low when we deployed some new infrastructure or technologies.”

StreamShark evaluated its infrastructure and app development options and elected to build its core platform on App Engine

(<https://cloud.google.com/appengine/>)—a fully

managed, serverless platform from Google Cloud (<https://cloud.google.com/>) that provides seamless scalability. The business's engineering team quickly built their skills and experience with the service, and StreamShark began using other Google Cloud services to seize the available efficiency gains. "We have a lot of smart people in our team and some very good account managers at Google who connect us to technical specialists when we need assistance," says Broberg.

"We really
appreciate the
Google Cloud
model and
prefer to run our
compute
workloads on
the platform."

—James Broberg, founder
& CEO, StreamShark

**Google Cloud cornerstone of
architecture**

Google Cloud now acts as the cornerstone of StreamShark's multi-cloud architecture. "While we have other providers in our mix, we lean heaviest on Google Cloud, particularly for compute," says Broberg. The business continues to use App Engine for its database and business layer, while Compute Engine (<https://cloud.google.com/compute/>) provides compute resources as a service and Cloud SQL (<https://cloud.google.com/sql/>) delivers relational database services for analytics and statistics. StreamShark also recently moved from the Cloud Datastore (<https://cloud.google.com/datastore/>) NoSQL database service to the Cloud Firestore (<https://firebase.google.com/docs/firestore>) cloud database to support interactive functionality such as chat.

StreamShark's constant search for ways to be more efficient without diminishing the client experience has led the business to Google Cloud virtual machines powered by NVIDIA GPUs. "Our business requires us to invest in infrastructure to ingest video content, so customers can send in video for transcoding and packetizing into a range of formats," explains Broberg. "That can be expensive and present scalability challenges when tens or hundreds of thousands of people send in video streams at the same time. The NVIDIA GPUs allow us to consolidate onto fewer, higher-powered pieces of infrastructure and delivered considerable efficiency and scalability gains."

Sustained-usage discounts available through Google Cloud have also delivered considerable benefits to StreamShark. "With Google Cloud, we don't have to

predict ahead of time how much we are going to use and pre-commit financially,” says Broberg. “We really appreciate the Google Cloud model and prefer to run our compute workloads on the platform.”

With Google Cloud, StreamShark is now living up to what Broberg describes as its reputation as “the guys you go to for a ‘can’t-fail video stream.’”

“When we execute a high-profile live stream for an important person, whether they are a CEO or head of state, we need to be confident the infrastructure is there and we have diversity in regions and availability zones, so any single failure does not cause

an issue for our
clients. Google
Cloud gives us
that assurance.”

—James Broberg, founder
& CEO, StreamShark

Diversity in regions and availability zones key

“When we execute a high-profile livestream for an important person, whether they are a CEO or head of state, we need to be confident the infrastructure is there and we have diversity in regions and availability zones, so any single failure does not cause an issue for our clients,” says Broberg.

“Google Cloud gives us that assurance.”

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—James Broberg, founder
& CEO, StreamShark

Google Cloud has also given Streamshark the elasticity to support spikes in demand that may occur at various stages—particularly the start—of a stream. “Google Cloud gives you the ability to scale out and deliver those big loads and peak audiences and scale that back when that load dissipates,” says Broberg. “It’s not uncommon for us to have hundreds of thousands or millions of concurrent users.”

Broberg is particularly excited by the role of Cloud Firestore (<https://cloud.google.com/firestore/>) in supporting live chat on video streams. “What we needed was a low-latency responsive experience for people participating in chat,” he says. “Their comments should appear immediately, and they should be able to have a dialogue with other users about the content they are watching.”

A proof of concept undertaken by StreamShark proved Cloud Firestore could solve a lot of the technical challenges involved in delivering high-speed, high-performance, interactive experiences.

AI and ML next step

StreamShark is now undertaking proofs of concept to determine how AI and machine learning technologies can provide better services to clients.

“We are pursuing AI and ML at the moment,” says Broberg. “Our motivation is to unlock the richness of the content we obtain with clients so we can provide more value and insight—for example, through the ability to search older videos using techniques to identify speakers.

“We’re also looking at text-to-speech and translation capabilities.”

Overall, he says, Google Cloud has provided the infrastructure that has enabled the business to meet the various challenging customer demands that have emerged since StreamShark’s inception. “It has been a very positive partnership for us,” concludes Broberg.



StreamShark CEO James Broberg

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